

Understanding Random Effects in Mixed Models

This handout introduces random effects using mountain gorillas. We begin with a single random effect, then build to nested and crossed random effects.

1. A Single Random Effect

Biological question

We collected multiple urine samples from the same gorillas:

- G01
 - Sample 1
 - Sample 2
 - Sample 3

- G02
 - Sample 1
 - Sample 2

Because the same gorilla appears multiple times, the observations are not independent.

Model

```
mod_random <- lmer(  
  cortisol ~ fruit_availability + respiratory_signs +  
    (1 | gorilla_id),  
  data = data  
)
```

Interpretation

The model allows each gorilla to have its own baseline cortisol level.

2. Nested Random Effects

Now suppose each gorilla belongs to one social group.

- Titus
 - G01
 - G02
 - G03

- Susa
 - G04
 - G05

— G06

Now there are two levels of grouping:

- Samples belong to gorillas.
- Gorillas belong to groups.

Because gorillas belong to one and only one group, the gorillas are nested within groups.

Model

```
mod_nested <- lmer(  
  cortisol ~ fruit_availability + respiratory_signs +  
    (1 | group/gorilla_id),  
  data = data  
)
```

Interpretation

This model allows:

- Groups to have different baseline cortisol levels.
- Gorillas within those groups to have different baseline cortisol levels.

Internally, R expands this model to:

```
(1 | group) + (1 | group:gorilla_id)
```

where **group:gorilla_id** creates unique identifiers such as Titus:G01 and Susa:G04.

3. Crossed Random Effects

Now imagine one female transfers groups.

Year 1

Titus

— G01

— G02

Susa

— G03

Year 2

Titus

— G02

Susa

— G01

— G03

Now G01 belongs to **two different groups** during the study. Gorillas are no longer permanently nested within groups.

Instead, each observation belongs to:

- One gorilla
- One group

Model

```
mod_crossed <- lmer(
  cortisol ~ fruit_availability + respiratory_signs +
    (1 | gorilla_id) +
    (1 | group),
  data = data
)
```

Interpretation

This model allows:

- Gorillas to have different baseline cortisol levels.
- Groups to have different baseline cortisol levels.

It does not assume that each gorilla remains in the same group throughout the study.

4. Why not use $(1 | \text{group/gorilla_id})$?

If G01 transfers groups, the nested model creates two different identities:

Titus:G01

Susa:G01

The model no longer treats these as the same gorilla. Instead, it treats them as two different gorilla-group combinations. That is appropriate only if each gorilla truly belongs to one group throughout the study.

5. Summary

Data structure	Model	Interpretation
Repeated measurements from the same gorilla	(1gorilla_id)	Gorillas have different baseline cortisol.
Gorillas stay in one group	$(1 \text{group/gorilla_id})$	Groups differ, and gorillas differ within groups.
Gorillas can move between groups	$(1 \text{gorilla_id}) + (1 \text{group})$	Gorillas differ, groups differ, and gorillas can change groups over time.

Take-home Message

Always choose a random-effects structure that reflects the biology of your study system.

- One random effect: repeated measurements from the same unit.
- Nested random effects: one level is contained within another.
- Crossed random effects: observations belong to two grouping variables that are not permanently linked.